Geometry & Visual-Spatial Problems AIME level

Curated for Andy — a mathematician who thinks in pictures. June 1, 2026.

Why these problems fit Andy's profile

Andy does his entire computation **internally** — no written steps, no whiteboard scaffolding — and communicates the result afterward by drawing. That single fact should drive what we put in front of him. These five problems were chosen because each one:

- **Collapses to one spatial insight.** The hard work is *seeing* the right picture (unfold it, bisect it, slice it). Once seen, the rest is a one-line mental calculation. This matches how he already works — he solved the IMO 2011 Windmill problem in his head, which is exactly this kind of "find the invariant" geometry.
- **Needs no written derivation.** There is no multi-page algebra to transcribe — a barrier for a child with fine-motor/writing impairment. The reasoning lives as a mental image, his native channel.
- **Outputs a single value.** The answer is one number (or one clean expression), which fits AAC / drawing-based communication far better than a written proof.
- **Rewards systemizing.** In Baron-Cohen's framing, Andy's gift is extreme systemizing — detecting lawful regularities. Each problem hinges on an *invariant*: distance is preserved when you unfold, a fold-line is a perpendicular bisector, tangent lengths from a vertex are equal, symmetry fixes the octagon and hexagon. These are pure structure, not memorized procedure.
- **Is sensory-light and self-contained.** Each can be held entirely in the mind's eye — no tools, no writing, no verbal exchange required to engage with it.

The throughline: **geometry that pays off mental visualization over written manipulation** — leaning into his strength while routing around the writing/verbal barriers, not testing them.

The Five Problems

1 · **The Spider and the Fly** (3D → unfold to 2D)

A box-shaped room is **30 long, 12 wide, 12 high**. A spider sits at one bottom corner; a fly at the diagonally opposite top corner. Find the **shortest distance crawling along the surfaces**.

- **Insight:** Don't reason in 3D — **unfold** the box flat so the path becomes a straight line, then use Pythagoras. Different unfoldings give different lengths; pick the smallest.
- Best unfolding: distance = $\sqrt{(30^2 + (12+12)^2)} = \sqrt{(900 + 576)} = \sqrt{1476} = 6\sqrt{41} \approx 38.4$.

- The "obvious" route (across floor, then up the wall) = 30 + 12 = 42 — longer. The unfolded straight line wins.

Answer: $6\sqrt{41}$

For Andy: the entire difficulty is a mental rotation — flattening a 3D box. That is a pure spatial act, no writing, and it's precisely the kind of dimensional re-seeing that visual thinkers do faster than symbol manipulators.

2 · The Folded Rectangle (2D paper-folding)

A rectangle is **4 wide, 3 tall**. Fold it so one corner lands exactly on the diagonally opposite corner. How long is the **crease**?

- **Insight:** Folding corner B onto corner D makes the crease the **perpendicular bisector of diagonal BD** — every crease point is equidistant from B and D.
- With A(0,0), B(4,0), C(4,3), D(0,3): crease runs (0.875, 0) → (3.125, 3).
- Length = $\sqrt{(2.25^2 + 3^2)} = \sqrt{14.0625} = 15/4 = 3.75$.

Answer: 15/4

For Andy: "equidistant from two points" is an invariant he can *see* as a mirror line. The fold is physical and visual; the answer is one fraction.

3 · The 13-14-15 Triangle (the famous "nice" triangle)

A triangle has sides **13, 14, 15**. Find its **inradius** and how the incircle splits each side at its tangent points.

- **Insight:** Tangent segments from each vertex are equal, and equal to (semiperimeter – opposite side).
- $s = 21$; Area (Heron) = $\sqrt{(21 \cdot 7 \cdot 6 \cdot 8)} = \sqrt{7056} = 84$; inradius $r = 84/21 =$ **4**.
- Tangent lengths: 21−14 = 7, 21−15 = 6, 21−13 = 8. So side 14 splits into **6 and 8**, side 15 into **7 and 8**, side 13 into **6 and 7** (each pair sums to its side — a built-in check).

Answer: r = 4; splits 6-7-8

For Andy: the equal-tangent-length structure is a clean symmetry rule. He can hold the whole figure as one picture and read off the integer splits — no algebra, just structure.

4 · Two Squares Overlapping (rotational symmetry)

Two identical squares of side **1** share a center; one is rotated **45°** (forming an 8-pointed star). Find the **area of the overlap** (a regular octagon).

- **Insight:** The rotated square is just the constraint $|x|+|y| \le \sqrt{2}/2$; its diagonals **slice off the four corners** of the first square.
- Overlap = 1 − (four corner triangles) = $2\sqrt{2} - 2 \approx 0.828$.

Answer: $2\sqrt{2} - 2$

For Andy: symmetry does all the work — recognizing four identical cut corners. It's a "see the whole pattern at once" problem, ideal for a systemizer.

5 · The Hexagon Hidden in a Cube (3D cross-section — the prettiest)

Slice a **unit cube** with a plane through the **midpoints of six edges**. The cross-section is a **regular hexagon**. Find its area.

- **Insight:** The six edge-midpoints are mutually equidistant. Adjacent vertices, e.g. (1, ½, 0) and (½, 1, 0), are $\sqrt{(¼+¼)} = \sqrt{2}/2$ apart — the hexagon's side.
- Regular hexagon area = $(3\sqrt{3}/2)\cdot s^2 = (3\sqrt{3}/2)\cdot(½) = 3\sqrt{3}/4 \approx 1.299$.

Answer: $3\sqrt{3}/4$

For Andy: the surprise — that slicing a cube yields a *perfect* hexagon — is a spatial revelation. Visualizing the slicing plane inside the cube is exactly the mental 3D manipulation he excels at.

Prepared by a3ai for Jacob · The common thread: redraw the picture (unfold it, bisect it, slice it) and the algebra nearly vanishes — built to reward Andy's internal, visual computation rather than written work.